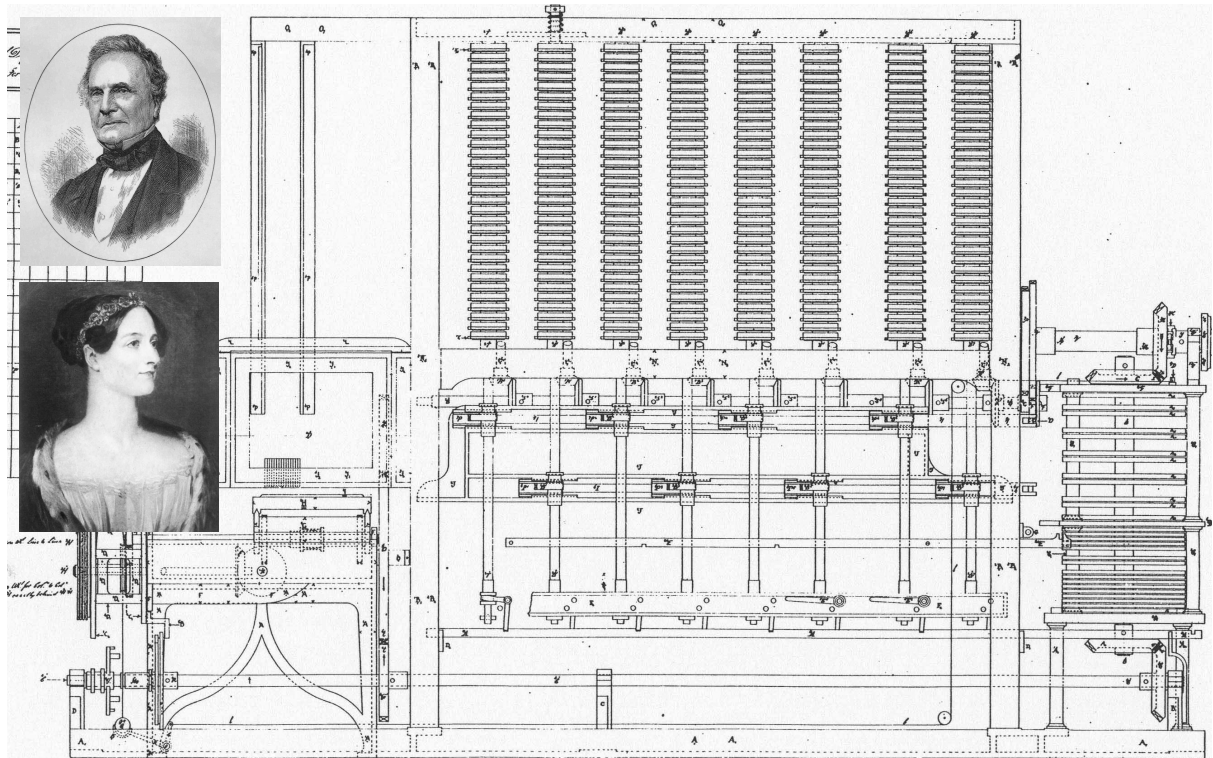


Charles Babbage, Ada Lovelace, and the Dawn of Computing

Instructor's Guide

Version 3.1



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Mark M. Meysenburg

Mark M. Meysenburg, 2016

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Chapter 1

Introduction

Welcome to *Charles Babbage, Ada Lovelace, and the Dawn of Computing*, a Reacting to the Past role-playing game centered on Charles Babbage and Ada Lovelace, pioneers of the modern computer who lived in Great Britain in the nineteenth century.

1.1 How to teach with reacting to the past

Reacting to the Past is a series of historical role-playing games. Students are given elaborate game books which place them in moments of historical controversy and intellectual ferment. The class becomes a public body of some sort; students, in role, become particular persons from the period, often as members of a faction. Their purpose is to advance a policy agenda and achieve their victory objectives. To do so, they will undertake research and write speeches and position papers; and they will also give formal speeches, participate in informal debates and negotiations, and otherwise work to win the game. After a few preparatory lectures, the game begins and the players are in charge; the instructor serves as adviser or “gamemaster.” Outcomes sometimes differ from the actual history; a post-mortem session at the end of the game sets the record straight.

1.1.1 Game set-up

Before the game begins, instructors must help students to understand the historical background. During the set-up period, students will read several different kinds of material:

- The game book, which includes historical information, rules and elements of the game, and essential documents; and
- Their roles, which describes the historical figures they will play in the game.

You may also assign primary and secondary sources outside the game book (perhaps including one or more accompanying books). Some texts are recommended in the annotated

bibliography that appears in the gamebook. Others may be suggested in the schedule section of this IM. If you want players to use these readings, they need to be tied in to the functioning of the game. Figure out how they tie into the game by relating them to specific roles and particular assignments. Characterize the set-up sessions as a brief introductory overview. Remind players that they should go back and reread these materials throughout the game. A second reading while in role will deepen their understanding. Remind players that players who have carefully read the materials and who know the rules of the game will invariably do better than those who rely on general impressions and uncertain recollections.

This Instructor's Guide provides prompts for leading discussions during these set-up sessions.

1.1.2 From instructor to gamemaster

Once the game begins, you become a Gamemaster (GM). During regular game sessions, this means you will often take a seat in the back of the room. While no longer in control, you may do any of the following:

- Pass notes to spur players to action;
- Announce the effects of actions taken inside the game on outside parties (e.g., neighboring countries) or the effects of outside events on game actions (e.g., a declaration of war); and
- Interrupt and redirect proceedings that have gone off track.

Much more of your work will occur outside of the classroom. Guide players by responding to their oral presentations and written work. Probably the best way you can help students is to provide nearly immediate feedback of both. Quick feedback is important because the game issues often shift rapidly.

In addition, it is quite likely that students (individually or in groups) will seek your counsel. Sometimes these consultations will involve confusion with the situation or game mechanics. Other times, they will involve students who are seeking some sort of in-game advantage. Thus, the more familiar you are with the game the better.

1.1.3 Student-centered classroom

Once the game begins, certain players preside over the class sessions. These presiding officers may be elected or appointed. The schedule section of this IG explains how this process works. Make sure that you have taken the necessary steps to select the first presiding officer before the game begins.

Presiding officers may act in a partisan fashion, speaking in support of particular interests, but they must observe basic standards of fairness. As a failsafe device, most Reacting games employ the "Podium Rule," which allows a player who has not been recognized to

approach the podium and wait for a chance to speak. Once at the podium, the player has the floor and must be heard.

Encourage students to avoid the colloquialisms and familiarities of today's college life. Never should the presiding officer, for example, open a session with the salutation, "Hi guys."

1.1.4 Role playing

Role sheets are extremely important to players. Given their unfamiliarity with role-playing and the chaotic and unpredictable nature of many of the class sessions, they will cling to them like life preservers in a stormy sea. Encourage them to keep their role sheets close and stress their confidential nature. Role sheets contain secrets!

It is unlikely that you will be able to master the contents of all the role sheets in this game – particularly the first time you use it. Consequently, encourage students to bring their role sheets along if they want to consult with you. Similarly, if you are corresponding with a student, pull their role sheet up on a screen so that you can consult it with ease.

Roles are often clumped into factions. This gives these players allies. In many games, one faction represents utopian theorists who seek to accommodate mankind to their intellectual visions; another faction represents social "realists" who seek to adapt these ideas to fit the obdurate shapes of human nature.

Remind faction members that in order to achieve their objectives, they will need the support of other students. They will never have the strength to prevail without allies. Consequently, collaboration and coalition-building are at the heart of every game. Along these lines, discourage them from resorting to violence in order to achieve their objectives. (Unless that is part of the learning objectives of this particular game). Remind these faction members that every game includes roles that are undecided (or "indeterminate") about certain issues. Similarly, encourage indeterminate roles by reminding them that they are the true kingmakers. Without their support, no faction can hope to prevail.

1.1.5 Liminality

Most games begin with some sort of "liminal moment." For example, *Threshold of Democracy* begins every session of the Athenian assembly with a pig sacrifice. These are odd rituals that are not unlike the cry of "play ball" at the beginning of a baseball game. They signal that the classroom has become a different place in which the students will be interacting in strange, unusual, and delightful ways. As the game continues, students may find that their liminality deepens.

1.1.6 Student discomfort

This sense of being immersed in a role may be particularly challenging to students charged with promoting worldviews that are antithetical to their own beliefs. If this causes discomfort, remind them that they are merely playing roles. Also remind them to direct their criticisms at one another's roles rather than one another as persons. (For example, you may need to intervene if someone repeatedly says, "Sally's argument is ridiculous." But encourage them to say, "Governor Winthrop's argument is ridiculous"). Similarly, remind students that it is inappropriate to trade on out-of-class relationships when asking for support within the game. ("Hey, you can't vote against me. We're both on the tennis team!")

Remind students to always assume, when spoken to by a fellow player – whether in class or out of class – that that person is speaking in role. Some roles may include elements of conspiracy or deceit. Such roles will cause some students stress, so you should encourage students to talk with you if they become uncomfortable with their roles. In the vast majority of cases, you will be able to talk them through their discomfort. To encourage these students make it clear that everyone is merely playing a role.

1.1.7 Victory

The challenges of achieving their victory objectives highly motivate many students even if the impact on their grades is insignificant.

1.1.8 Assignments

In general, RTTP games require several distinct but interrelated activities:

- **Reading:** This standard academic work is carried on more purposefully in a Reacting course, since what students read is put to immediate use.
- **Research and Writing:** The exact writing requirements depend on you, but in most cases students will be writing to persuade others (particularly the indeterminates).
- **Public Speaking and Debate:** Expect most of your students to deliver at least one formal speech from the podium.
- **Strategizing:** Communication among students is a pervasive feature of Reacting games. Encourage them to continue the game outside of class. You may want facilitate this by organizing their initial faction meetings perhaps during a regular class meeting.

Some game-specific variations on these requirements are described in the Assignments section of the gamebook, but for the most part, the particular structure of these assignments is up to you. Tailor the game to fit your learning objectives by consulting the suggestions in the Assignments section of this IG.

1.1.9 Schedule

Similarly, this IG includes a number of sample schedules. They should help you to fit the game to a variety of formats as well as learning objectives.

1.1.10 Ahistorical outcomes

Every game includes the potential for ahistorical outcomes. These fall within a “plausibility corridor” of possible counterfactual outcomes that have been designed by the author. If it is important for you to retain historical verisimilitude you may want to keep this corridor narrow. You can do this by nudging players to take certain actions or through *deus ex machina* interventions. In either case, it is usually best to do this outside of regular game sessions. Otherwise, students begin to feel as if they are your puppets.

Alternatively, if your learning objectives feature leadership, writing, and speaking you may want to release these controls. As you balance between encouraging students and staying true to the history, you may find yourself in a dilemma. For example, if a weak student who rarely speaks makes a presentation that is riddled with historical errors, should you immediately correct those errors publicly, which will ensure that the class learns the correct history, or should you wait, let the mistakes go uncorrected, and build the student’s confidence? Alternatively, what if an irrepressible student manages to cobble together an implausible coalition? Should you jump into the fray by forcefully reminding each faction of its purposes, or do you let it play out? This requires subtle judgment on complicated matters of content, student psychology, and pedagogy. That is to say, you must be a good teacher.

1.1.11 Debriefing

Every game ends with at least one session dedicated to debriefing. Comparing the historical record with student experiences is often an excellent pedagogical exercise, which helps students to understand historical causation and contingency. If nothing else, it provides you with an opportunity to set the record straight. In addition, this session allows students to exit the game. They put aside their game names, reveal their secrets, and disclose any skullduggery. Encourage them to tell all – it is important for them to put the conflicts between their roles behind them.

1.1.12 Modifications

Once you are familiar with the workings of the game, feel free to modify the game as you see fit (to go out on your own, in readings, written assignments, etc.). It’s your game now.

1.2 Brief introduction to the game

Charles Babbage, Ada Lovelace, and the Dawn of Computing takes place in early nineteenth-century Britain, focusing on the calculating engines designed by Charles Babbage. The central intellectual collisions in the game concern the conflict between imagination and reason, the nature of science and scientists (are they talented, wealthy amateurs, or is science a profession?) and whether and to what degree science and engineering projects should be subsidized by the government. The main question in the game is whether or not Charles Babbage should be awarded funds from the British government for the development of his Difference Engine (an automated calculator capable of automatically creating, typesetting, and printing mathematical tables) and / or Analytical Engine (a true proto-computer), during the early to mid-1800s.

The Engines, at a minimum, would have “mechanized number,” allowing rapid, automatic, error-free production of mathematical tables to be used in navigation, astronomy, and other applications. Ada Lovelace, regarded as the world’s first computer programmer, could envision applications of the Analytical Engine beyond mere mathematics, however. In the notes to her English translation of Luigi Federico Menabrea’s article *Sketch of the Analytical Engine Invented By Charles Babbage, Esq.* [Men43], Lovelace realized that the Engine could operate on symbols in an abstract manner, that it “might act upon other things besides number... the Engine might compose elaborate and scientific pieces of music of any degree of complexity or extent.” She also foresaw applications in business, drawing graphics, and more. In addition, the precision mechanical engineering work done on the engines had profound impacts on the state of engineering in the UK and around the world. The mind boggles at the prospects had an Analytical Engine ever actually been built. What would the consequences of a steam-powered, mechanical computer have been in Victorian England? One can easily get lost in the “steam-punk” possibilities of a 100-year head start in computing technology.

Players in the game may belong to one of two factions, or may be indeterminate. The pro-Babbage faction wants the government to provide funding adequate for the construction of one or both of Babbage’s Engines. The anti-Babbage faction wants the opposite – no funding for either Engine, either because they do not feel the Engines are that important, or because they do not feel the government should provide funding for such projects. Indeterminates have no set position on this issue, and are open to persuasion by the factions.

The game covers a broad span of time, from 1828 through 1846. There are six game sessions and three optional labs, briefly described in Table 1.1.

After a few class sessions to set up the game, you as the gamemaster will assign every student to a role in the game. As mentioned above, the roles are divided into three groups: the pro-Babbage faction, the anti-Babbage faction, and the indeterminates. The players have different victory objectives, which you will find in their role sheets. Be aware that some characters have victory conditions that are not tied to the performance of their faction. In most cases, however, players can win by persuading the indeterminates serving as PM to vote in favor or not in favor of funding Babbage’s engines during GS 3 and 6.

Table 1.1: Overview of game sessions

Session (year) / location	Presiding character	Theme
GS 1 (1828) / 1 Dorset Street	Babbage	Babbage's party
Lab 1	n/a	Finite differences
GS 2 (1828) / Mechanic's Institute	Lardner	Public lectures 1
Lab 2	n/a	Difference Engine
GS 3 (1828) / Royal Society	Gilbert	Difference Engine
GS 4 (1830) / Royal Society	Gilbert	Celebrating the RS RS presidency
Lab 3	n/a	Analytical Engine
GS 5 (1846) / Mechanic's Institute	Brunel	Public lectures 2
GS 6 (1846) / Royal Society	RS president	Analytical Engine

1.3 Learning objectives

By the end of the game, students will be able, in both written and spoken work, to do the following:

- Recognize the major men of science and politicians of Georgian and Regency Britain
- Identify the scientific, engineering, personality, and political issues that prevented Babbage's computing machines from being funded by the government and fully developed
- Identify the major components and architecture of Babbage's Difference Engine and Analytical Engine
- Use the method of finite differences and Taylor series expansions to produce approximate values for polynomials and other mathematical functions such as sine, cosine, and the natural logarithm
- Employ expanded research skills to find multiple resources that support their arguments
- Sharpen verbal expression, writing skills, and organization of ideas through the process of composing rhetorically appropriate speeches and traditional essays
- Differentiate between possible historical motivations
- Demonstrate how individual characters might react to concrete historical situations
- Devise and present logical arguments, supported by evidence
- Evaluate and appreciate classical historical and literary texts
- Analyze the relationship between fact and contingency, theory, and practice
- Criticize opposing viewpoints and defend their own

- Appraise distant cultures and transcend personal cultural constraints
- Collaborate with classmates as a member of a team
- Create community through teamwork (both inside and outside of class)
- Develop leadership skills, as members and leaders of teams

The players must also engage with questions such as:

- Should to government support scientific and engineering projects? If so, to what extent? And, must the projects have immediate perceived benefits for the government and / or society, or are pure science projects allowed?
- What societal changes were wrought by the industrial revolution?
- How did the engineering techniques developed during Babbage's computing engine projects filter out and benefit mechanical engineering practice in the UK?
- How did the nature of science, and scientists, change during the early nineteenth century?
- Which is more valuable: reason or imagination?

1.4 Key concepts

Key concepts for this game include the nature of science, the societal value of science and engineering, *laissez-faire* economic thought, the expanding role of government spending, the role of women in science and in society, the tension between reason and imagination, and the professionalization of science.

Chapter 2

Game setup

This section contains information to help you plan your version of the game and what you need to know before your class starts playing.

2.1 Before you start: logistics and resources

As a Reacting instructor, you should make several decisions well before the class containing the game begins, and several more before the game itself begins. There are also some logistics to consider. These issues are covered below.

2.1.1 Choosing a class schedule

First and foremost, you should choose a class schedule and whether to play the full version of the game. The model schedules section, below, has more information regarding these options. Make this decision based on the number of class sessions you wish to devote to the game, the amount of time you have available for each class session (e.g., 50 or 75 minutes), the number of students that will be playing the game, and the amount of time you want to devote to each speech, to open debate, and so on.

2.1.2 Selecting a classroom

Since students will be speaking to the class throughout the game, the classroom you use for the game should have a podium, or at least a designated location that can act as one. Ideally, the classroom you choose should be larger than you would normally require for the number of students enrolled, so that students can move around, confer in their factions, and so on. Reacting games typically become quite spirited (noisy), and so if possible, a classroom that is some distance away (in time or space) from other classes would be appropriate.

2.1.3 Preparing the syllabus

It is easy to directly map the schedule of sessions you select on to your syllabus. It is wise to allow for three sessions of pre-game setup, and at least one session for the postmortem after the game is over. The author or other instructors that have played the game would be happy to share sample syllabi with you, as well; check the Facebook Reacting to the Past Faculty Lounge to contact like-minded colleagues.

2.1.4 Expectations for length of speeches and writing assignments

As you are preparing your syllabus, you should decide what expectations you have for the lengths of the scheduled speeches students will present, and the length for their written work. The role sheets assign formal speeches and papers, and give default length expectations. Base your decision on the lengths of the class periods in your game, the level of sophistication of your students (i.e., is this a first-year seminar, or a 300 / 400-level course?), your grading scheme, and so on.

2.1.5 Library reserves

You may wish to add books from the included Bibliography to your library's reserve list. In particular, *An Illustrated Book of Bad Arguments* [Alm13] or similar resource will be helpful for Lardner and Airy, but not required for everyone in the class. In fact, the other roles should not be introduced to the book, lest they catch on to what Airy and Lardner are trying to accomplish.

2.1.6 Learning management system or other online presence

During the game, the students must read and refer to each other's papers, in both their written work and in their speeches. This is best done via a learning management system (such as Blackboard), a real-time messaging system (such as Slack), or some other electronic facility. Whatever method you choose, please make sure that only your current students have access. A publicly-accessible archive of the papers from a class makes it easy for other students to plagiarize papers and speeches for their version of the game. Also, please do not distribute electronic versions of the game materials to students, so that copyrighted materials do not make their way on to the Internet.

2.1.7 Preparing as gamemaster

If you have never played a Reacting to the Past game, you should! Seriously consider playing a condensed version of one or more games at a Reacting workshop. Running a

game is much easier if you have played it at least one time. For a schedule of upcoming Reacting workshops and institutes, including which games are being played, refer to the Reacting Consortium's Website: www.reactingsonsortiumlibrary.org. You can also see various videos of Reacting games on resources like YouTube.

To prepare for the game, you should first read through the gamebook. You may choose to skip the core texts section of the gamebook the first time through. After that, you should read this Instructor's Guide. Finally, you should read the role sheets for the roles you will be assigning.

2.1.8 Materials needed for the game

Before the game setup sessions take place, you should decide which student will play each role involved in your version of the game; see the role allocation table (Table 2.4, below) for guidance on assigning roles. Then, you should print out the appropriate role sheets (contained in the Instructor's Materials document). Distribute the role sheets to your students no later than the last day of the game setup sessions. Emphasize to your students that their role sheets are private and should not be shared with anyone else in the class.

Before GS 1, prepare a method to randomly select an indeterminate to be Prime Minister for GS 1, 2, 3, 4, and 5. Each eligible indeterminate will be PM in one of the first five game sessions, but the order of their service should remain a mystery to them. Randomly select and announce the PM of the day at the beginning of each of the first five game sessions. The author has used slips of paper drawn from a Victorian top hat to determine the PM of the day.

At the end of the setup sessions, you may wish to give a faction quiz to assess the students on their retention of the pre-game reading (and also to help make sure they read the pre-game reading). A sample faction quiz is included in the handouts section of the Instructor's Materials document. The quiz is taken in groups, by faction. The pro-Babbage faction members take the quiz as one unit, as do the anti-Babbage faction members; the indeterminates work together as a faction for the purposes of the quiz. The faction with the highest score on the faction quiz is awarded a draw from the *power card* (PC) deck. If the indeterminates score highest on the quiz, draw lots to determine which indeterminate gets to draw. The gamemaster should award the faction quiz PC at the beginning of GS 1. If the indeterminates won the quiz, bring a 10-sided die to class to determine the character that receives the PC.

Before the game begins, make sure you have a complete deck of playing cards (52 cards plus two jokers) to serve as the PCs in the game. PCs are the currency of the game; the more PCs accumulated by a player or faction, the more likely that player or faction will win the game. PCs are mapped to the deck of cards as shown in Table 2.1.

The powers assigned to the cards are described in the gamebook. In addition to the cards shown in Table 2.1, the jokers represent the power of Celebrity. These cards, given to Ada when the role sheets are distributed are unique to her. More details regarding how the jokers can be used are shown in Ada's role sheet.

Table 2.1: Power cards

Card	Power	Card	Power	Card	Power	Card	Power
A ♣	Laissez faire	A ♦	Intervention	A ♥	Prescience	A ♠	Muddle
K ♣	Swing Riot	K ♦	Invigorate	K ♥	Volte-Face	K ♠	Volte-Face
Q ♣	Cholera	Q ♦	Tar & Vinegar	Q ♥	Shove	Q ♠	Shove
J ♣	Excess	J ♦	Excess	J ♥	Exflunctify	J ♠	Topsy
10 ♣	Nudge	10 ♦	Nudge	10 ♥	Nudge	10 ♠	Nudge
9 ♣	Nudge	9 ♦	Nudge	9 ♥	Nudge	9 ♠	Nudge
8 ♣	Nudge	8 ♦	Nudge	8 ♥	Nudge	8 ♠	Nudge
7 ♣	Nudge	7 ♦	Nudge	7 ♥	Nudge	7 ♠	Nudge
6 ♣	Nudge	6 ♦	Nudge	6 ♥	Nudge	6 ♠	Nudge
5 ♣	Nudge	5 ♦	Nudge	5 ♥	Nudge	5 ♠	Nudge
4 ♣	Nudge	4 ♦	Nudge	4 ♥	Nudge	4 ♠	Nudge
3 ♣	Nudge	3 ♦	Nudge	3 ♥	Nudge	3 ♠	Nudge
2 ♣	Nudge	2 ♦	Nudge	2 ♥	Nudge	2 ♠	Nudge

There are two types of PCs: *nudges* and *powers*. Nudges are used to exert influence in three ways:

1. Nudges can influence the PM when he makes a decision regarding funding for Bab-bage's engines at the end of GS 3 and GS 6.
2. Nudges can be used to help steer the selection of the indeterminate who will serve as PM in GS 6.
3. Nudges can be used to purchase the services of the Ornamental Hermit, if he becomes a part of the game after another character has died of cholera. See the Ornamental Hermit role sheet for more information.

The rules for how nudges and powers are used are outlined in the gamebook.

Certain characters begin the game with one or more PCs, so those PCs should be distributed with their respective role sheets. The characters receiving initial PCs are shown in Table 2.2. Distribute the PCs in such a way that no one but the receiving character can tell which cards have been distributed. The author has distributed PCs in sealed envelopes along with the respective role sheets.

Also, you may wish to print and distribute name tags and / or name tents for each of the characters in your game. A special, "traveling" name tent could be used to indicate which of the indeterminates is currently serving as Prime Minister.

At the end of each game session, you will conduct a poll of the indeterminates; this is termed the "persuasiveness metric." The persuasiveness metric is used to determine the indeterminates opinion of who made the most persuasive presentation during the game session, and the winner receives a PC. Conduct the poll via a secret ballot. To facilitate this, you could:

Table 2.2: Initial PCs – distribute before GS 1

Character	Initial PCs
The Ghost of Byron	1 nudge
Gilbert	1 nudge
Grey	1 nudge
Lovelace	2 jokers
Peel	1 nudge
Robinson	1 nudge
Russell	1 nudge
Sussex	1 nudge
Wellington	1 nudge

- use blank slips of paper handed to each indeterminate;
- use slips of paper pre-printed with the names of those scheduled to present during the session, so the indeterminates only need to check their vote for the best presentation (printable versions of these are included in the Instructor’s Materials); or
- use an electronic polling mechanism of some sort

Likewise, at the end of GS 6, all of the players with franchise to vote (refer to the table in Part Three of the gamebook to see who has franchise) will vote on the PM who they thought made the best presentation during his PM speech. The winner of this “election” will be the winning indeterminate in the game. You will need some method of voting for this part of the game; options like those just stated for the persuasiveness metric are feasible alternatives here. Printable voting slips are included in the Instructor’s Materials.

Finally, at the end of GS 6, all of the players will vote on the supremacy of reason or imagination. This will decide if either King or Byron achieves their victory conditions. You will need some method of voting for this part of the game; options like those described above are feasible. Printable voting slips are included in the Instructor’s Materials.

2.2 Model schedule

The recommended schedule of six game sessions is shown in the gamebook. The three labs are optional and may be omitted if you desire.

2.3 Roles and factions

Deciding which roles to include in the game is a function of how many students you have in the class and the characteristics of each student. Assigning roles is a major responsibility for the gamemaster. Some gamemasters carefully assign roles based on their students’

Table 2.3: MOS membership

Character	MOS member?	Character	MOS member?
Adams	Y	Herschel	Y
Airy	Y	Ornamental Hermit	
Babbage	Y	King	
Brunel	Y	Lardner	Y
Buckland	Y	Lovelace	Y
Byron’s Ghost		Peel	
Clement	Y	Robinson	
Crosse	Y	Russell	
Faraday	Y	Sabine	Y
Frend		Somerville	Y
The Fury		Sussex	
Gilbert	Y	Swing	
Grey		Wellington	

strengths, weaknesses, and personalities; some assign roles randomly. There is a spectrum of options for deciding who gets which role between these two extremes.

2.3.1 Brief description of roles and factions

Each of the roles in the game is briefly described below. The primary speaking responsibility for each role is summarized, and any secret elements pertaining to a role are noted. Each member of the “men of science” (MOS) community has a secret language responsibility, described in a separate handout. The members of the MOS community are shown in Table 2.3.

2.3.1.1 Pro-Babbage faction

The pro-Babbage faction is a collection of men of science, engineers, and personal friends of Charles Babbage. This group is committed to the development of Babbage’s computing engines, recognizing how important they will be. The faction members also recognize that developing the Difference Engine or the Analytical Engine will be expensive, beyond the capacity of private financing.

John Couch Adams ~~The~~ John Couch (pronounced “cooch”) Adams is a British astronomer and mathematician, best known for his co-discovery with Urbain Le Verrier of the planet Neptune in 1846. Adams is important to his faction, because he has an era-appropriate, concrete example of how the Analytical Engine could have been beneficial to Britain. He will make the case that his calculations regarding the location of Neptune would have been completed even sooner had one of Babbage’s engines been

available. **Secret:** Adams can win the game, separately from his faction, if the Royal Society passes a resolution that he should be considered the discoverer of Neptune.

Charles Babbage FRS Charles Babbage is a British polymath – a person whose expertise spans a significant number of different subject areas – and the inventor of two types of calculating machines. First, Babbage invented the Difference Engine, a device capable of automatically calculating and printing certain types of mathematical tables, and then the Analytical Engine, a true proto-computer. Babbage presides over his Dorset Street party in GS 1. **Secret:** Babbage is almost a saboteur in his own faction. He has a responsibility that is counter to his own best interests. During GS 4, he will attack the Royal Society, based on *Reflections on the Decline of Science in England, and on Some of its Causes* [Bab30].

Isambard Kingdom (I. K.) Brunel FRS I. K. Brunel is perhaps the most accomplished mechanical and civil engineer in Britain during the time of our game. Brunel, a close personal friend of Charles Babbage, is responsible for completing the Thames Tunnel project, for building the Great Western Railway, and for the construction of the steamship S. S. Great Western. Brunel presides over the second set of public lectures at the Mechanic’s Institute, during GS 5. In addition, Brunel speaks during GS 5 on the success of the Thames Tunnel project, extending its success to make the case for government funding of other engineering projects, specifically, Babbage’s Difference or Analytical Engines.

Joseph Clement Joseph Clement is the extremely capable toolmaker, draftsman, engineer, and industrialist hired by Charles Babbage to build his Difference Engine. **Secret:** Clement will defect from the pro-Babbage faction in GS 5, joining the anti-Babbage faction and earning a draw from the PC deck for the faction. Clement will give a presentation during GS 5 detailing why he defected.

Andrew Crosse Andrew Crosse is an amateur man of science, best known for his experiments with electricity. Mary Shelly, the author of the novel *Frankenstein*, is an acquaintance, and her husband Percy Bysshe Shelley attended one of Crosse’s lectures on atmospheric electricity in 1814; this may or may not have helped to influence later versions of the famous novel. Crosse makes a presentation during GS 5 on the new concept of interchangeable parts, and how the concept might be applied to Babbage’s Difference Engine.

Michael Faraday Michael Faraday is an extremely influential English man of science; of all the characters in this game, his influence is the greatest in our 21st-century world. His work focuses on chemistry and electricity. Faraday makes a presentation during GS 3, introducing a resolution recommending that the government should fund work on Babbage’s Difference Engine.

John Frederick William Herschel FRS John Frederick William Herschel is another British polymath, and close friend of Charles Babbage. He and Babbage, among others, formed the Analytical Society while undergraduate students at Cambridge. Herschel studied mathematics, astronomy, chemistry, botany, and more. Herschel is the leader of the pro-Babbage faction. Herschel also has a responsibility to introduce

the Difference Engine fragment at Babbage’s Dorset Street party during GS 1. In addition, Herschel is the pro-Babbage candidate for PRS.

Dionysius Lardner FRS Dionysius Lardner is a member of the Royal Society of London and a professor at the University of London. Lardner is a scientific populist and the first scientific journalist. Lardner presides over the first set of lectures at the Mechanic’s Institute during GS 2. In addition, Lardner will speak during GS 2, on the value of the Difference Engine. Lardner’s speech will focus on the perceived economic benefit of the Difference Engine, specifically the harm that could occur due to errors in the navigational tables used by mariners. One way he will achieve this is to provide the tables Sabine will use in his navigation problem. **Secret:** There are two versions of the tables: one with errors and one without. Lardner and Herschel should work together to decide who receives each version during GS 2, and not tell anyone that some of the tables have errors until after the erroneous results have been found. Also, Lardner can win the game separately from his faction, if he is able to intentionally use and document at least two bad arguments (straw man, false dilemma, slippery slope, etc.) during the game, without getting caught. Lardner does not know it, but the anti-Babbage faction leader, Airy, should be especially on the alert for bad arguments.

Augusta Ada King-Noel, Countess of Lovelace Augusta Ada King-Noel, Countess of Lovelace, is the only legitimate daughter of the “mad, bad, and dangerous to know” romantic poet, Lord Byron. Lovelace is a close friend of Charles Babbage, and in 1843, she translates an article on Babbage’s Analytical Engine by Luigi Federico Menabrea, and adds her own voluminous notes to the translation. The notes contain what is now considered the world’s first computer program. Lovelace will make a presentation on the Analytical Engine at the second set of Mechanic’s Institute lectures, during GS 5. In her speech, she will briefly explain how the machine would work, and argue why constructing the machine is vital. Her presentation will try to show the benefits of the Analytical Engine apart from mere mathematical calculation and tabulation. Other characters may try to leverage Ada’s celebrity by reciting Byronic stanzas to her in return for a draw from the PC deck. Ada receives the two Jokers from the PC deck at the beginning of the game; see her role sheet for how she can use these “Power of Celebrity” cards.

Mary Somerville Mary Fairfax Greig Somerville is a Scottish polymath. She is very unusual for the time, having published several very influential papers and books. Somerville served as one of Ada Lovelace’s tutors, and introduced her to Charles Babbage at one of his Dorset Street parties. Somerville will work together with Frend in GS 1 to teach the other characters in the game how to use a slide rule and mathematical tables to solve problems.

2.3.1.2 Anti-Babbage faction

The anti-Babbage faction is a collection of men of science, clergy, and political figures. This group is against the British government paying any more money for the development

of Babbage's computing engines. Anti-Babbage faction members have different reasons for opposing the engines: some do not think the engines will be valuable or necessary for the advancement of science in Britain; some think the engines are potentially valuable, but do not think that projects such as this should be supported by the government; and some are generally against the increasing mechanization of British society, as it forces working people from their jobs.

George Biddell Airy FRS George Biddell Airy is an English mathematician and, as of 1835, the Astronomer Royal. Airy, holder of the Lucasian Professorship of Mathematics at Cambridge before Charles Babbage assumed the chair in 1828, is the leader of the anti-Babbage faction. Airy will make a presentation during GS 6 regarding the priority of the discovery of the planet Neptune, arguing that Adams is overstating not only his claim to be the discoverer of the planet but also how valuable a computing engine would have been to the project. **Secret:** Airy is the game's designated curmudgeon, assigned to dislike just about everyone and everything. Part of this assignment is to call out bad arguments (straw man, false dilemma, slippery slope, etc.) when they are used by other characters. In this way, he is Lardner's foe, since Lardner is assigned to make bad arguments as part of his victory conditions. Neither Airy nor Lardner know anything of these aspects of the other's roles, however.

Reverend William Buckland DD FRS The Reverend Doctor William Buckland, DD FRS, is an English clergyman, theologian, geologist, and paleontologist. Buckland published the first full account of a fossilized dinosaur, which he termed *Megalosaurus*. During GS 3, Buckland will make a speech introducing a resolution against the funding of Babbage's Difference Engine to the Royal Society. The speech will summarize all the reasons why the anti-Babbage faction believes that spending government money on the Difference Engine is unwise. **Secret:** To play off Buckland's zo'ophagy, he should bring animal crackers, gummy worms, or other animal-shaped snacks to each game session. He should nonchalantly eat the snacks during the game sessions, without drawing overt attention to the practice.

William Frend William Frend is an English Unitarian clergyman, political radical, and author of the algebra textbook, *Principles of Algebra* [Fre96]. Frend, along with Mary Somerville, served as one of Ada Lovelace's mathematics tutors. Frend will work with Somerville in GS 1 to teach the other characters in the game how to use a slide rule and mathematical tables to solve problems.

Davies Gilbert PRS Davies Gilbert is a British botanist, geologist, mathematician, civil servant, politician, author, and from 1827 through 1830, president of the Royal Society of London. As PRS, Gilbert will preside over game sessions 3 and 4. During GS 4, Gilbert will make a speech about the history and accomplishments of the Royal Society.

Edward Sabine FRS Edward Sabine, FRS, is an Irish military man, geologist, and astronomer. He served as the on-board astronomer on expeditions to attempt to discover the Northwest Passage under Sir John Ross in 1818 and again under Sir William Edward Parry in 1819 and 1820. Sabine will make a speech during GS 2, teaching

the other characters how to solve a more complicated mathematical problem involving trigonometry and Pythagoras' Theorem. **Secret:** Lardner will supply the tables that will be needed to solve Sabine's problem; Sabine does not know that some of the tables will contain errors that prevent arriving at the correct answer for Sabine's challenge.

Prince Augustus Frederick, Duke of Sussex, KG KT GCB FRS FRSA Prince Augustus Frederick, Duke of Sussex, is the sixth son of “mad” King George III, who reigned from 1760 until 1820. Although not a man of science, Sussex does have intellectual pursuits, chiefly in biblical studies and the Hebrew language. Sussex is the anti-Babbage faction's candidate for PRS in GS 4. Sussex will make a speech on the principles of *laissez-faire* economic principles during GS 2.

Captain Swing Captain Swing represents the spirit and ideas of the farmers who participated in the “Swing Riots” across England in 1830 and 1831, starting fires and breaking threshing machines, protesting for increased wages, against restrictive Poor Laws, and against the increasing use of threshing machines in agriculture. Several of the threatening letters sent by the protesters were signed with the *nom de guerre* “Captain Swing.” Swing will make a presentation during GS 2 at the Mechanic's Institute on the plight of Britain's farmers. These issues led to the Swing Riots. Swing will extrapolate from the way threshing machines put farmers out of work to draw unemployment implications regarding Babbage's Difference Engine.

2.3.1.3 Indeterminates

In this game, the characters that will serve in the role of Prime Minister are indeterminate, and there are several other indeterminates that do not serve as PM. None of these characters have ~~no~~ pre-decided position regarding Babbage's engines – although each character may have some general personal or political leanings that may inform their opinions on the subject. Each PM will make a speech when he takes office, outlining the priorities that will guide his government. Some of the other indeterminates will make speeches, but not all of them.

Each eligible indeterminate (Grey, Peel, Robinson, Russell, and Wellington) will serve as PM once during the first five game sessions. You should randomly choose the PM before each game session. This means that the students assigned as these indeterminates should all be ready to give their PM speech in GS 1, since no one knows when they will serve.

The indeterminates are polled by the GM at the end of every game session for their vote for the most persuasive presentation made during the session. The winning presenter receives a PC.

At the end of GS 3 and 6, the current PM will make the decision whether to fund one of Babbage's machines, within the constraints of the nudge stacks described in Part Three of the gamebook. Factions and / or individuals spend nudges before the PM's decision, adding each nudge to one of the stacks assigned for use during that game session. The PM may also have one or more nudges that he can spend at his discretion. The stack with

the most nudges is the decision the PM must make. If there is a tie, the PM can decide between the tied stacks at his discretion.

The Ghost of Lord Byron Byron’s Ghost is the voice of the romantic virtue of imagination. He will speak behind the scenes with other characters in the game to convince them that imagination is more important than reason. Byron’s Ghost is not eligible to be PM.

The Fury The Fury is an amalgam of the women employed by Lady Anne Isabella Noel Byron to supervise the education and upbringing of her teenage daughter, Ada. The Fury is responsible for making a presentation during GS 1 on the proper behavior of ladies and gentlemen, and then for making sure that everyone conforms to these standards while the game is running. She also wishes to thwart other characters from taking advantage of Ada’s celebrity, by detecting and calling out errors they make when they recite Byron’s verses to Ada. The Fury is not eligible to be PM.

Lord Charles Grey, 2nd Earl Grey, KG PC Lord Charles Grey, 2nd Earl Grey is a Whig MP, and the namesake of Earl Grey tea. Grey will serve as PM at least once.

The Ornamental Hermit The Ornamental Hermit enters the game if a character dies from cholera (due to the “Cholera” PC). The dead character comes back as the Ornamental Hermit. The hermit is responsible for dressing oddly and reciting melancholy poetry for other characters during the social time in each game session. The Ornamental Hermit is not eligible to be PM.

Doctor William King William King is Ada Lovelace’s personal physician and mentor. In this game, King will speak during GS 1 on the value of reason over the romantic virtue of imagination. Doctor William King is not eligible to be PM.

Sir Robert Peel, 2nd Baronet, FRS PC Sir Robert Peel, 2nd Baronet is a Tory MP. Peel will serve as PM at least once.

Lord Frederick John Robinson, 1st Earl of Ripon, FRS PC Lord Frederick John Robinson, 1st Earl of Ripon is a Tory MP. Robinson will serve as PM at least once.

Lord John Russell, 1st Earl Russell, KG GCMG PC Lord John Russell, 1st Earl Russell is a Whig MP. Russell will serve as PM at least once.

Field Marshal Arthur Wellesley, 1st Duke of Wellington, KG GCB GCH PC Field Marshal Arthur Wellesley, 1st Duke of Wellington is the biggest hero of the French Wars and Tory MP. Wellington will serve as PM at least once.

2.3.2 Role allocation table

The role allocation table below (Table 2.4) indicates which characters should be included in the game, depending on class size. The Ornamental Hermit may appear in any game, or not at all, depending on whether someone dies of cholera during the game.

2.3.3 Brief description of extra roles

As this game undergoes further development, additional roles will be added to increase the total number of students that can play the game.

2.4 Rules and procedures

Most of the rules for the game are outlined in the gamebook and the role sheets for the characters. More detail regarding the rules are listed below.

2.4.1 Winning the game

It is important for both the gamemaster and the students to realize that there will be multiple winners in the game. Each character has victory conditions, stated in their role sheets. Most characters win only if their faction wins: pro-Babbage faction members win if the government decides, on both opportunities, to fund Babbage's computing machines, while the anti-Babbage faction members win if the government declines to fund the machines on both opportunities. If the decisions are split, then the faction with the largest number of remaining unused nudges wins. If both factions have the same number of unused nudges, the GM will roll a die to determine the winning faction.

Other characters have an extra way to win apart from the victory of their faction. For example, Lardner wins the game if the pro-Babbage faction wins, or if he successfully makes at least two bad arguments (straw man, slippery slope, false dilemma, etc.) without being caught by Airy or another character. MP Indeterminates win in a different way – by vote of the characters with franchise at the end of GS 6. The Fury wins if she prevents at least two characters from exploiting Ada's celebrity or if she catches, calls out and documents at least five lapses in the rules of polite society. Byron's Ghost and William King win or lose based on the outcome of the final vote on imagination vs. reason.

2.4.2 Power cards

Power Cards (PCs, mapped to a deck of playing cards as shown in Table 2.1) are the currency of the game. Characters and factions can earn PCs in various ways, such as:

- By scoring the highest on the faction quiz at the end of the game setup sessions
- By wearing the best period-appropriate costume during a game session, as judged by the GM
- By winning the Georgian party game that takes place during Babbage's party, in GS 1
- By making the best presentation during a game session, as judged by the GM

- Based on the outcome of the Royal Society vote in GS 3, to recommend or not recommend funding the Difference Engine
- For the anti-Babbage faction only, passing any Royal Society resolution that is unfavorable to Babbage
- If the Royal Society passes a resolution favorable to a faction or an individual character
- As a result of a Persuasiveness Metric vote at the end of a game session
- By convincing Countess Lovelace to allow you to draw a PC from the deck
- When Clement defects, the anti-Babbage faction earns an PC
- At the beginning of each game session, one of the characters in the game will be randomly selected; this character gets to draw a PC
- Byron's Ghost draws an extra PC at the beginning of GS 1
- By virtue of their fame, wealth, or political power, Byron's Ghost, Gilbert, Sussex, Grey, Peel, Robinson, Russell, and Wellesley all begin the game with one nudge PC each

Nudges are spent at the end of GS 3 and 6. Each nudge spent is placed on one of the decision stacks for the game session (the stacks are described in Part Three of the gamebook). No one has to spend their nudges; they can be saved to be more impactful for later decisions. During the nudge spending part of GS 3 and 6, individuals spend their nudges first, then factions, and finally the current PM. The decision stack with the most nudges in it is the decision the PM must make. If two stacks are tied for the maximum number of nudges, then the PM can decide between the tied stacks at his discretion.

2.4.3 Persuasiveness metric

At the end of each game session, the indeterminates vote on the most persuasive presentation of the session. Possible mechanisms for conducting the poll are listed in the logistics section, above. For most game sessions, the winner of the persuasiveness metric vote is awarded PC draw. The winner of the PC should be announced at the beginning of the next game session, along with the PC draw awarded by the GM for his or her best presentation. For the game sessions where the PM makes funding decisions (GS 3 and 6), the gamemaster should conduct the poll of the indeterminates before the funding decision, and award all PCs (including those given by the GM) before the nudge spending phase.

2.4.4 Winning PM vote

Near the end of GS 6, all of the characters with franchise to vote (see the table in Part Three of the gamebook) vote for the PM who made the best PM speech during the game. The winner of this vote is the winning MP indeterminate for the game. If two or more

PMs are tied for first place, then all of the tied PMs meet their victory conditions. The winning PM need not be announced until the postmortem session.

2.4.5 Faction defection

During GS 5, Clement defects from the pro-Babbage faction to the anti-Babbage faction. He makes a speech during the game session, outlining the reasons why he left the employ of Babbage, and his defection earns the anti-Babbage faction a PC. This defection reflects Clement's actual departure from the Difference Engine project, which dealt the enterprise a fatal blow.

As gamemaster, you should make sure that Clement understands that he should not reveal his intentions to defect to anyone before the event happens in GS 5. Also, you should prevent any other characters from trying to defect.

2.4.6 Who can speak

During the sessions that do not take place at the Royal Society (Babbage's party in GS 1, and the public lectures at the Mechanic's Institute in GS 2 and 5), any character may speak publicly. During sessions that take place at the Royal Society (GS 3, 4, and 6), however, only current members of the Royal Society are allowed to speak publicly from the podium or ask questions of those speaking at the podium. A table in Part Three of the gamebook shows who is a FRS, and when that status begins. For example, Airy was not yet elected FRS, and so he may not speak publicly during GS 3 and 4. Lovelace and Somerville are excluded because they are women. Clement, as a working man, is also excluded. Some non-MOS characters are FRS (Peel and Robinson), while other characters that are influential MOS are not FRS (Adams and Crosse). The presiding officer for the sessions at the Royal Society (Gilbert in GS 3 and 4, and then the winner of the PRS vote for GS 6) should be looking out for this, and as gamemaster, you should help them enforce this rule. Color coded nametags might be helpful in this regard.

The characters that cannot speak publicly during Royal Society meetings may confer with their faction members and other characters; the restriction applies to asking questions of others that are speaking and publicly addressing the whole body from the podium.

2.4.7 Podium rule

The podium rule states that any character who wishes to speak during an open discussion period should approach the podium. Once queued up at the podium, the presiding officer for the game session must allow the character to speak. The presiding officers for the game sessions, shown in Table 2.5, belong to one of the factions. Therefore, it is possible – even though their role sheets prohibit the behavior – that the presiding officer tries to ignore characters from the opposing faction when they approach the podium. As gamemaster,

Table 2.5: Presiding officers and PRS

GS	Presiding officer	PRS
1	Babbage	Gilbert
2	Lardner	Gilbert
3	Gilbert	Gilbert
4	Gilbert	Gilbert
5	Brunel	PRS
6	PRS	PRS

you cannot allow this favoritism. During the game, if you detect this behavior from the presiding officer, you should send them a note telling them to let everyone who wishes to speak. If the behavior continues, the threat or actual act of awarding PCs to the relevant factions should prevent a recurrence of this kind of behavior.

Characters are encouraged to ask questions of presenters, at the discretion of the presiding officer, but they should not make speeches to the class without approaching the podium.

If Brunel is not in your game due to a low number of students, then you as GM can preside over GS 5, or some other pro-Babbage faction member could serve in that capacity.

2.5 Reading and writing assignments

This section briefly summarizes the reading and writing assignments the students playing the game are expected to complete.

2.5.1 Readings

The readings assigned for this game are voluminous, and can be difficult for students at first. The style of prose used in writing in the early 1800s United Kingdom may be initially off-putting to students. In addition, some of the more mathematical content (functions, polynomials, finite differences, Taylor series approximations, etc.) may be especially problematical for students without a mathematical affinity. As gamemaster / instructor, you may wish to prepare students in advance for the readings. At a minimum, students should be aware that a single read-through of the core texts in the gamebook will not be sufficient to compete in the game.

The lab sessions, while optional, can help students solidify their understanding of the readings, by forcing them to work in a “hands on” manner with the material. This argues for including the labs if your syllabus allows the time to do so.

2.5.2 Writing assignments

As in all courses, you should clearly state at the outset the amount of writing expected of the students in the game. Most players in the game have two writing assignments. The first assignment is a written version of the speech the character is assigned to make. These papers should be posted to the document-sharing system you are using for the class before the start of the game session scheduled for the presentation.

The second assignment for most characters is actually a series of five shorter papers. In these papers, characters in the factions write and privately submit to the GM a few paragraphs on their impressions of each PM speech. Characters can use these writings to inform their vote for the winning PM at the end of GS 6.

MP Indeterminates have a different second writing assignment, also a series of shorter papers. They will write a few paragraphs explaining who they voted for in the persuasiveness metric portion of each game session, and why. These papers are to be submitted privately to the GM before the beginning of the next game session.

Lardner has an optional third writing assignment, related to his duty to intentionally make bad arguments during the debates in the game. Lardner should write a short paper detailing which bad argument techniques he employed, along with their definitions, and when each was used. This paper should be submitted privately to the gamemaster before the beginning of the post-mortem session. Airy has a similar paper, turned in if he has successfully detected, called out, and documented three bad argument techniques.

The assignments for each character are listed in the Assignments section of their role sheet.

2.5.3 The need for independent research

Remind students early and often that they are expected to go beyond the materials included in their role sheets and the core texts included in the gamebook when they prepare their speeches and written work. In addition to other works of history, students should also refer to material written by other students in the game.

2.5.4 Grading

Grading students in a Reacting game involves several components: reading comprehension quizzes (the faction quiz given at the end of the setup phase), writing, speaking, and class participation. Feel free to evaluate your students in the manner you are most comfortable with. Advice on grading options, including sample rubrics, can be found in the Reacting Faculty Lounge Facebook group. To gain access to this faculty-only resource, contact the Reacting Consortium at reacting@barndard.edu.

Many Reacting instructors include some sort of tangible reward for students who meet their victory conditions in the game. For example, one might add a one-half grade level increase

in winning students' participation scores. In a specifications grading model, winners could receive an extra "do-over" certificate, allowing them to resubmit a failed assignment from another portion of the course. Generally speaking, these rewards for winning are not terribly consequential in terms of the student's final grade in the course, but provide psychological motivation for students to become more fully engaged in the game.

Chapter 3

Managing the game

What follows are step-by-step guidelines on how to manage the elements in this Reacting game. Your main task as gamemaster is to serve as guide and cheerleader. Students need to be encouraged to plunge into the proceedings, take risks, and master difficult ideas and complicated materials. You should meet with each character in the game individually, before the game starts, to make sure each person understands the expectations associated with his or her role. Encourage students to do more than the required research and seek a more in-depth understanding of their characters beyond the information provided in their role sheets.

Also be sure to grade student papers swiftly: if they are going off track, they need to be alerted as soon as possible. Generally speaking, the goal is to return graded work by the start of the next game session.

3.1 Setup sessions

The setup, or pre-game, sessions are designed to help students prepare for the game. Some recommended activities are:

1. Discuss the RTTP pedagogy in general. An “Introduction to Reacting” handout is available from the faculty-only library on the Reacting Website

(www.reactingsortiumlibrary.org).

Obviously, this pedagogy may be quite different from what students have done before, and so you may try to proactively address concerns they have.

2. Do some public speaking exercises to warm students up to the type of interaction they are expected to do during a Reacting class. Lily Lamboy has created a series of two-minute videos with associated pair practicing. Links to these videos are also available on the Reacting Website.

3. Provide one or more introductory lectures / discussions on the historical context of the game (ending with the year 1828). The Gamebook has a vignette that sets the scene, and excerpts from some of the primary texts involved in the game. Your pre-game lectures / discussions do not need to cover everything in depth – the students are supposed to be doing in-depth research as the game progresses. Rather, you are giving an outline and setting the context.
4. Assign roles and distribute role sheets to the students in the class. Remind students (again!) of two things:
 - (a) First, each student should meet with the gamemaster individually before the beginning of GS 1, to make sure they understand what they are required to do.
 - (b) Secondly, remind students that their role sheets are for their eyes only, and should not be shared with anyone else. Please do not share the role sheets electronically, to cut down on the likelihood of them becoming publicly accessible on the Internet for future students playing the game.

Certain roles may also receive other handouts at this time. For example, each MOS member should receive a copy of the MOS Secret Language handout, and Babbage should receive his instructions on the GS 1 party game at this time. In addition, Byron's Ghost, Gilbert, Sussex, Grey, Peel, Robinson, Russell, and Wellesley all start the game with a nudge; these can be distributed with the respective role sheets. Likewise, Lovelace receives her two jokers when she receives her role sheet.

5. Conduct the reading comprehension quiz. The quiz is taken in groups, by faction; for the purposes of the quiz, the indeterminates are a faction. The faction that scores the highest on the reading comprehension quiz receives a draw from the PC deck at the beginning of GS 1.
6. Faction meetings. It is wise to allow at least twenty minutes for the pro-Babbage and anti-Babbage factions to meet and organize. This should take place after all of the other pre-game activities, on the last class session before the game begins.

3.2 Materials to bring for every game session

Bring the following materials with you for every game session:

- Name tags with pins or lanyards, or name tents to place on desks. Students must refer to each other by character name during the game; name tags or tents help students meet this requirement, especially early in the game. It is wise to collect the tags or tents at the end of each game session so they are not lost between sessions. You may wish to color code the tags / tents to indicate which characters have FRS status.
- For the first five game sessions, bring some method of randomly selecting the PM of the day. Slips of paper drawn from a top hat do nicely. Draw the name of one of the

remaining indeterminates and then notify the class who will serve as PM before the game session starts.

- A special name tag or name tent to identify the current Prime Minister.
- This Instructor's Guide, or an abbreviated version of the schedule, so you know who is supposed to speak during each game session.
- Time cards or some other method of keeping track of the time taken for each presentation.
- Note paper and / or rubrics for tracking evaluating students during the game session.
- A smaller notepad or slips of paper for passing notes to students. On occasion you may wish to pass notes to characters in order to keep the game moving.
- The deck of power cards. Once powers have been played, you should keep the discard pile separate from the unused pile, and take care not to shuffle the unused cards unless compelled to do so by one of the powers.

3.3 GS 1: Babbage's party (1828)

GS 1, Babbage's party, takes place in 1828 at Babbage's residence, 1 Dorset Street, in London. Babbage is the presiding officer for the session. Robinson is the PM.

Extra items to bring or arrange for:

- A mechanism for randomly choosing an indeterminate, if the indeterminates won the faction quiz (see below)
- *I Love my Love with an A* party game handouts, one for every character except Charles Babbage
- Slide rules and table of cosines for Frend and Somerville to use in their mathematics lesson. The author assembles one of the circular slide rules per player in the game, and brings one copy of the cosine table per player as well
- A mechanism for recording forfeits during the party game (see below)
- A mechanism for polling the indeterminates for the GS 1 Persuasiveness Metric; cut-out voting slips are included in the Instructor's Materials

If the indeterminates won the faction quiz at the end of the setup phase, have an appropriate method of randomization to determine which indeterminate receives the associated PC.

Begin class by randomly selecting and announcing which of the indeterminates is serving as PM for the session.

The session begins with a lecture from the Fury on the rules for behavior in polite society.

Then, we play the Georgian-era party game, adapted from the 1825 book *Winter Evening Passtimes; or, The Merry-Maker's Companion* [Rev25]. Babbage serves as the “master of the revels,” i.e., he runs the game. Babbage also makes sure that everyone plays fair and that no one cheats; cheaters are assigned “forfeits.” Bring copies of the game instructions handout for each player except Babbage; he should have received his version of the game instructions when the role sheets were distributed. Be prepared to track forfeits assigned during the game, perhaps on the whiteboard or via a projected spreadsheet.

The player with the fewest number of forfeits will receive a PC at the beginning of GS 2. If more than one player is tied for the fewest number of forfeits, you should break the tie via a die roll, drawing a name from a hat, or some other randomized method.

After the revelry of *I Love my Love with an A*, the remaining presentations scheduled for the game session take place. Babbage should call the following characters to make their presentations, in this order:

1. The current PM makes his PM speech
2. Frend and Somerville teach how to use a slide rule and mathematical tables to solve problems
3. Doctor William King speaks on why reason should be valued over imagination
4. Herschel speaks about the method of finite differences and the Difference Engine fragment Babbage displays in his home

Make sure to reserve time at the end of the game session to conduct the Persuasiveness Metric poll of the indeterminates. The indeterminates vote on who they thought made the most persuasive presentation during the game session. The winner of this Persuasiveness Metric will receive a PC at the beginning of GS 2.

3.4 Lab 1: The method of finite differences [Optional]

Lab 1 teaches the students about two key mathematical concepts that were key to the Difference Engine: the method of finite differences, and the concept of Taylor Series approximations. This lab is optional.

Students may complete the lab individually, in pairs, or in factions (if in factions, the indeterminates can work together).

Extra items to bring / prepare for:

- Enough lab handouts for the number of individuals or groups that will be completing the lab
- If you wish to demonstrate / run code related to the lab, make sure the classroom computer has the required software (the Java Development Kit, Python, etc.)

There are optional programming assignments with the lab. Each of the programming assignments requires a different level of programming skill to complete; consider the background of your students when assigning these pieces. If you do not assign the programming components, you may wish to demonstrate how one would solve the problems in code.

3.5 GS 2: Public lectures at the Mechanic's Institute (1828)

GS 2, the first series of public lectures at the London Mechanic's Institute, takes place in 1828. Dionysius Lardner is the presiding officer.

Extra items to bring / prepare for:

- A die or other method for breaking the tie if more than one person won the GS 1 party game (see below)
- A mechanism for polling the indeterminates for the GS 2 Persuasiveness Metric; cut-out voting slips are included in the Instructor's Materials

Begin class by randomly selecting and announcing which of the indeterminates is serving as PM for the session. Each PM serves once in the first five game session, so be sure to exclude the PM from GS 1 from selection.

If there was a tie for the fewest number of forfeits in the GS 1 party game, bring some method of breaking the tie to class at the start of GS 2. Break the tie in the presence of the students, and then award the PC to the winning character. Then, award a PC to the winner of the Persuasiveness Metric poll, and another to the player you, as the gamemaster, thought made the best GS 1 presentation.

After all the PCs have been awarded, Lardner should take over the class and manage the following presentations:

1. Social time: a small amount of time for players to mingle, strategize, etc.
2. The current PM makes his PM speech
3. Sussex speaks on the benefits of *laissez faire* economic policies
4. Sabine teaches the players how to solve a navigational problem using trigonometry and Pythagoras' Theorem, using mathematical tables provided by Lardner. Some of the tables have errors, so not everyone will be able to solve the problem correctly
5. Lardner himself speaks on the Difference Engine and the importance of correct tables
6. Captain Swing speaks on the plight of Britain's agricultural workers and the dangers of automation

If time remains after all of the scheduled presentations, the floor is open for debate. Lardner must use the podium rule to recognize those who wish to speak.

At the end of the game session conduct the Persuasiveness Metric poll of the indeterminates. The indeterminates vote on who they thought made the most persuasive presentation during the game session. The winner of this Persuasiveness Metric will receive an PC at the beginning of GS 3.

3.6 Lab 2: The Difference Engine [Optional]

Lab 2 teaches the students more about the workings of the Difference Engine. Students may complete the lab individually, in pairs, or in factions (if in factions, the indeterminates can work together).

Extra items to bring / prepare for:

- Enough lab handouts for the number of individuals or groups that will be completing the lab
- Each person or faction completing the lab will require access to a computer with an internet connection and a modern Web browser. If the students are expected to bring their own computers for the lab, make sure to notify them ahead of time
- It is wise to verify connectivity, by navigating to the simulator at <http://dunx1.irt.drexel.edu/~cp676/DiffEngine.html>, before the lab begins

3.7 GS 3: Royal Society recommendation on the Difference Engine (1828)

GS 3 takes place at the Royal Society of London in 1828. During the session, the Royal will decide whether to recommend that the Treasury fund the Difference Engine or not. Davies Gilbert is the presiding officer.

Extra items to bring / prepare:

- A mechanism for FRS voting on the Difference Engine resolution(s). A hand vote may be taken and recorded, or slips of paper could be used
- A mechanism for polling the indeterminates for the GS 3 Persuasiveness Metric; cut-out voting slips are included in the Instructor's Materials

Begin class by randomly selecting and announcing which of the remaining indeterminates is serving as PM for the session. Make sure you remove the PMs from GS 1 and 2 from consideration. Then, award PCs for the winner of the GS 2 Persuasiveness Metric, and for the player you thought made the best presentation.

After the PCs have been awarded, Gilbert takes over and manages the following presentations:

1. Social time: a small amount of time for players to mingle, strategize, etc.
2. The current PM makes his speech
3. Buckland introduces a resolution against funding the Difference Engine
4. Faraday introduces a resolution supporting funding the Difference Engine

Depending on enrollment, your game may not include Faraday and / or Buckland. If not, you can assign someone from each faction to introduce the respective resolution, or you can allow the factions to determine who should do the speaking. Sample resolutions can be found in the Instructor's Materials.

After both resolutions have been introduced, Gilbert should open up the floor to debate. He must use the podium rule to make sure that everyone who wishes to be heard receives an opportunity to speak. As game master, keep an eye on the time for the open debate, reserving enough time for the following activities.

First, after the open debate, the Fellows of the Royal Society vote on which resolution to approve (pro-Babbage or anti-Babbage). The winning faction immediately receives an PC.

Then, conduct the Persuasiveness Metric poll of the indeterminates. The indeterminates vote on who they thought made the most persuasive presentation during the game session. The winner of this Persuasiveness Metric will immediately receive an PC.

You, as the gamemaster, should also immediately award your best presentation PC.

Next, nudge spending takes place. You, as the gamemaster, should keep track of the spending with the nudge spending, in two stacks: one to fund the Difference Engine, and one to not fund it. The fund stack starts with two nudges already in it (which need not be actual cards). In this process, individuals, factions, and the PM spend their nudges by distributing them across the stacks. Individuals spend their nudges first, then factions, and finally the PM. Each party in the spending process may spend as many or as few of the nudges they control as they choose.

Finally, the PM decision regarding funding for the Difference Engine takes place. Once the spending is complete, the stack that has the most nudges in it is the decision the PM must make. If there is a tie, the PM is allowed to make the decision as he sees fit.

3.8 GS 4: Celebrating the Royal Society (1830)

GS 4 also takes place at the Royal Society of London, but two years after the events of GS 3. As far as the students are concerned, the session is to be a celebration of the Royal Society. In fact, the primary purpose of the session is to serve as a platform where Babbage can bring forth his criticisms of the Royal, as outlined in his book, *Reflections on the Decline*

of *Science in England, and on Some of its Causes* [Bab30]. Davies Gilbert is the presiding officer.

Extra items to bring / prepare:

- A mechanism for voting on PRS. Each FRS gets one vote.
- A mechanism for polling the indeterminates for the GS 4 Persuasiveness Metric; cut-out voting slips are included in the Instructor's Materials

Begin class by randomly selecting and announcing which of the remaining indeterminates is serving as PM for the session. Make sure you remove the PMs from GS 1, 2, and 3 from consideration.

After the PM has been selected, Gilbert takes over and manages the following presentations:

1. Social time: a small amount of time for players to mingle, strategize, etc.
2. The current PM makes his speech
3. Gilbert speaks on the history and accomplishments of the Royal Society
4. Babbage speaks on his criticisms of the Royal Society. Note that the subject of his speech should come as a surprise to everyone except Babbage
5. The factions nominate their candidates for next PRS. The pro-Babbage faction should nominate Herschel, while the anti-Babbage faction should nominate Sussex.
6. Open floor for debate

After the open debate, all FRS should vote on the next PRS. The winning character becomes the next President of the Royal Society, and will preside over GS 6.

At the end of the game session conduct the Persuasiveness Metric poll of the indeterminates. The indeterminates vote on who they thought made the most persuasive presentation during the game session. The winner of this Persuasiveness Metric will receive an IP at the beginning of GS 5.

3.9 Lab 3: The Analytical Engine [Optional]

Lab 3 teaches the students more about the workings of the Analytical Engine. Students may complete the lab individually, in pairs, or in factions (if in factions, the indeterminates can work together).

Extra items to bring / prepare for:

- Enough lab handouts for the number of individuals or groups that will be completing the lab

- Each person or faction completing the lab will require access to a computer with an Internet connection and a modern Web browser. If the students are expected to bring their own computers for the lab, make sure to notify them ahead of time
- It is wise to verify connectivity, by navigating to the simulator at
<https://www.fourmilab.ch/babbage/cards.html>,
before the lab begins

3.10 GS 5: Public lectures at the Mechanic's Institute (1846)

GS 5 takes place at the London Mechanic's Institute in the year 1846, 16 years since the previous session. During this session, the focus of the game will turn from Babbage's Difference Engine to the Analytical Engine. GS 5 is also when Clement makes his speech outlining his difficulties with Babbage, and then defects from the pro-Babbage to the anti-Babbage faction.

This is the last session to take place outside the Royal Society, so you may wish to remind non-FRS characters to take advantage of the session to make any podium speeches they need to make before the game concludes.

Extra items to bring / prepare:

- A mechanism for polling the indeterminates for the GS 5 Persuasiveness Metric; cut-out voting slips are included in the Instructor's Materials

Before the session starts, read or distribute the handout, "Significant Events Between 1830 (GS 4) and 1846 (GS 5)." This indicates that, regardless of the outcome of the vote in GS 3, the government has continued to fund the Difference Engine.

At the beginning of class, announce the PM for the day, i.e., the one remaining indeterminate who has not yet served as PM. Then, you can award the PCs from the previous game session.

After the PCs have been awarded, Brunel takes over and manages the following presentations:

1. Social time: a small amount of time for players to mingle, strategize, etc.
2. The current PM makes his speech
3. Clement makes his speech, explaining why he is parting ways with Babbage. After his speech, he defects and becomes a member of the anti-Babbage faction. Award the anti-Babbage faction a PC for this defection.
4. Brunel speaks on the Thames Tunnel, illustrating the value of government funding for large-scale engineering projects

5. Lovelace speaks on the Analytical Engine
6. Crosse speaks on the power of interchangeable parts

After the debate is finished, conduct the Persuasiveness Metric poll of the indeterminates. The indeterminates vote on who they thought made the post persuasive presentation during the game session. The winner of the Persuasiveness Metric will receive a PC at the beginning of the next session.

3.11 GS 6: Royal Society discussions (1846)

In GS 6, the action returns to the Royal Society. This session is devoted to debate regarding the Analytical Engine. One of the MPs will be chosen at random to serve as PM for the session. Nudges can be used to influence this election; see the gamebook for the rules on how to conduct this.

Extra items to bring / prepare:

- A mechanism for polling the indeterminates for the GS 6 Persuasiveness Metric; cut-out voting slips are included in the Instructor's Materials
- A mechanism for determining the winning PM, by vote of the players with franchise. Cut-out voting slips are included in the Instructor's Materials
- A mechanism for voting on reason or imagination. Cut-out voting slips are include in the Instructor's Materials

At the beginning of class, the PRS assumes control of the Royal Society, and manages the following presentations:

1. Social time: a small amount of time for players to mingle, strategize, etc.
2. Adams speaks on the discovery of the planet Neptune, arguing that, had an Analytical Engine been available, his calculations could have been completed more quickly, and Britain would have been first to discover the planet
3. Airy also speaks on the discovery of Neptune, countering Adams' arguments regarding the supposed utility of the Analytical Engine
4. Open floor for any resolutions and debate

At the end of the game session, conduct the Persuasiveness Metric poll of the indeterminates. The indeterminates vote on who they thought made the most persuasive presentation during the game session. The winner of the Persuasiveness Metric will immediately receive a PC. You should also award your PC for best presentation immediately.

Next, nudge spending takes place. You, as the gamemaster, should keep track of the spending with the nudge spending, in two stacks: one to fund the Analytical Engine, and one to not fund it. The fund stack starts with two nudges already in it (which need not be actual cards). In this process, individuals, factions, and the PM spend their nudges by

distributing them across the stacks. Individuals spend their nudges first, then factions, and finally the PM. Each party in the spending process may spend as many or as few of the nudges they control as they choose.

Next, the players with franchise (see the table in part three of the Gamebook) should vote for the PM they felt made the best PM speech during game sessions one through five. Do not announce the winner of the vote until the post mortem, *after* the PM has made his funding decision.

Next, all players vote on their preference for either imagination or reason. This will determine if Byron's Ghost or William King wins.

Finally, the PM decision regarding funding for the Analytical Engine takes place. Once the spending is complete, the stack that has the most nudges in it is the decision the PM must make. If there is a tie, the PM is allowed to make the decision as he sees fit.

3.12 Postmortem sessions

After the game is over, you should spend one or more class sessions conducting a post-mortem of the game. During the postmortem, you announce the winners of the game, uncover the various secrets that were in the game, ask the students to do some reflection regarding the game's content, discuss what really happened in history regarding Babbage and his engines, and so forth. The secrets from the game are summarized here, and an overview of the historical material for the postmortem is contained in the next section.

The secrets from the game are:

- The MOS secret language shared by the men of science in the game
- Adams' secret victory condition: he wins if the Royal Society passes a resolution regarding his priority in discovering Neptune
- Babbage's counterproductive responsibilities
- Lardner's bad argument responsibility
- Airy: curmudgeon and bad argument detector
- Buckland's zöophagy and snacking
- Clement's defection, although this is not a secret after GS 5

Chapter 4

Debriefing

4.1 Instructions for determining victory

Victory conditions for each character are included in each role sheet, and are straightforward. Faction members win the game if their faction wins; pro-Babbage if Babbage's engines are funded in two of the three opportunities, and anti-Babbage if the engines are not funded in two of the three opportunities.

The PM voted as having the best PM speech at the end of GS 6 is the winning MP indeterminate. The Fury wins if she prevents at least two characters from leveraging Lovelace's celebrity or if she can document at least five cases in the game where she reminded others to mind the rules of polite society. Byron's Ghost or King wins depending on the outcome of the GS 6 imagination vs. reason vote. The Ornamental Hermit wins if someone pays at least one nudge to employ him.

Lardner wins, apart from his faction, if he is able to use and document at least two bad arguments in his speeches, without getting caught.

Adams wins, apart from his faction, if he is able to convince the Royal Society to pass a resolution supporting his claim for priority in the discovery of the planet Neptune.

4.2 Instructions on exiting the game

Announce all the winners, and reveal secrets, as covered in the post-mortem section above. Then, you may wish to cover some or all of what really happened (see below), and compare and contrast that with the outcome of the game.

4.3 What really happened

As outlined in the Historical Background section of the Gamebook, Babbage first conceived of the idea for the Difference Engine in 1820 or 1821. A six-wheel prototype was completed, and on July 3rd, 1822, Babbage announced the project in a letter to Sir Humphrey Davy, the president of the Royal Society. Davy sent a copy of the letter to Sir Robert Peel, who was serving as Home Secretary at the time.

Later in 1823, Babbage met with Viscount Goderich (Robinson), who was serving as the Chancellor of the Exchequer, seeking government support for the Difference Engine project. Goderich promised funding, but was very vague; the expectation was that the Engine would be completed in three years. No notes or minutes of the meeting were taken, so the specifics of what was really promised are lost. Babbage thought Goderich had made a more open-ended promise of support. Babbage receives £1,500 for the project.

In 1827, Babbage traveled on the Continent, chiefly to get over the sudden loss of his wife and youngest child. While touring Europe, Babbage visited several manufacturing facilities, looking for ideas to incorporate into the Difference Engine project.

By 1828, the Difference Engine was still not complete, due in large part to Babbage changing the plans for the machine frequently. Babbage requested more funding from the government to complete the project. The Chancellor of the Exchequer, Henry Gouldburn, asked the Royal Society for its recommendation. This is the decision made by the Royal Society in session three of the game. The Royal recommended that the government should continue to fund the project, and Babbage would ultimately receive approximately £17,000 more. Babbage, wanting to move work on the project closer to his residence, began to convert his carriage house to a fireproof workshop.

In 1830, Babbage published *Reflections on the Decline of Science in England, and on Some of its Causes* [, babbage1830] alienating many colleagues in the Royal Society.

In 1833, Babbage and Clement had a falling out, which for all practical purposes spelled the end for the Difference Engine project. Babbage and Clement disagreed about payments owed to Clement; Babbage thought Clement was overcharging for his work, and deliberately working slowly. Clement, of course, disagreed. He refused to move to the new workshop without expensive concessions from Babbage. Babbage refused these, causing Clement to quit. It was the custom at the time that an engineer was the owner of all the tools he developed to work on a particular project, and so when Clement quit, he kept all of the machine tools that were designed to create the parts of the Difference Engine. He also kept the plans Babbage had drafted for the Engine. After an extended period, Babbage got his plans back, but not the tools. Babbage was left with two problems: first, how to replace the very competent and experienced Clement with another engineer to build the Difference Engine. Secondly, even if Babbage found someone capable of building the Engine, the new engineer would have to recreate tooling to make the parts of the Engine, and this was cost prohibitive.

Also in 1833, Babbage realized he could design a simpler, and yet much more powerful, calculating engine. This became known as the Analytical Engine.

In 1834, Babbage asked the Chancellor of the Exchequer for guidance: should the government provide more funding for the Difference Engine, switch to funding the Analytical Engine project instead, or cancel the calculating engine project entirely. There was no answer from the government for eight years.

In 1842, Sir Robert Peel, serving as Prime Minister, canceled the calculating engine project.

In 1843, Luigi Federico Meabrea's article on the Analytical Engine, *Sketch of the Analytical Engine Invented by Charles Babbage, Esq.* [Men43], with Lovelace's voluminous notes, was published. This is the only contemporary, published description of the Analytical Engine.

In 1848, Babbage made one more request for Analytical Engine funding from the British Government, but was denied.

4.4 What happened afterward

Ada Lovelace died of uterine cancer on November 27, 1852. There is still some controversy between historians regarding the role she played in Babbage's story. Was she truly the world's first programmer, or merely an advanced student who happened to publish some of Babbage's ideas? Recently, while researching for her graphic novel *The Thrilling Adventures of Lovelace and Babbage: With Interesting & Curious Anecdotes of Celebrated and Distinguished Characters, Fully Illustrating a Variety of Instructive and Amusing Scenes; as Performed within and without the Remarkable Difference Engine, Embellished with Portraits and Scientific Diagrams* [Pad15], Sydney Padua came across an account of Babbage's visit to the United States in the 1860s. This account makes it clear that Babbage thought the world of Ada, and considered her to be a mathematician of the first order. It does seem clear that Ada saw the true potential of the Analytical Engine – how it might work with words or music or art – more clearly than any other contemporary.

In Sweden, a printer named Georg Scheutz, along with his son Edvard, inspired by Lardner's reports of Babbage's Difference Engine, began the design and construction of his own engine. He began in 1834, and after many years of work, considerable personal expense, and only a very small amount of government funding, their machine was completed in 1853. The Scheutz machine handled fifteen-digit numbers and four orders of differences, and included a printing mechanism. In 1854 Goerg and Edvard Scheutz brought their engine (which was considerably smaller than Babbage's engine would have been) to England, where it was received very positively by the Royal Society, and, in fact, by Babbage himself. The Scheutz engine worked, but was not as robust as Babbage's design; it was prone to jamming and to breaking down. Nevertheless, the British government purchased an exact copy of the Scheutz engine in 1863, to be used in producing books of insurance tables. The original engine was purchased by the Dudley Observatory in Albany, New York.

Babbage continued to work on designs for his Analytical Engine until his death in 1871. He was always thinking of new ways to improve the engine, and on how to economize the potential production of the machine. As time went on, however, Babbage became known as

something of a crackpot in London. In particular, Babbage had a long-running feud with street musicians, petitioning the local government on more than one occasion to prohibit them from playing outside his residence.

Even though it was never built, the architecture of Babbage's Analytical Engine is very much like that of modern computers. The Analytical Engine's mill corresponds to the modern central processing unit (CPU), while the Engine's store corresponds to the volatile memory (RAM). The Engine's punch-cards correspond to the non-volatile storage (hard drive, SD card, etc.). Even the instruction set that the Engine would have used is immediately familiar to the assembly language modern computer programmers learn in their schooling.

It is clear that Babbage's work was ahead of his time, and as is often the case, his designs for the Analytical Engine were largely forgotten by the start of the twentieth century. It is possible that Alan Turing, another Cambridge man, knew of Babbage's work and was inspired by it when he worked on his "Bombe" at Bletchley Park to break the German Enigma code during the second World War.

Computing did not truly come into its own until electronic components replaced mechanical parts, starting in the mid 1930s. In Germany, Konrad Zuse built a working mechanical computer in 1936, while the Atanasoff-Berry Computer, the first electronic digital computer, was built at Iowa State University in 1937; this marked the true birth of the modern computing age.

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